

BRAC University  
CSE340: Computer Architecture  
Quiz#3

Duration : 20 Minutes

Total Marks : 20      Set: Abacus

**ID:**

**Name:**

**Sec:**

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1. Draw a simplified datapath with control unit that can process *only* the following codes:  
LD X20, 20(X22)  
SD X22, 10(X20) [12]
  2. Does the ALU control utilize instruction bits 30 and 14-12 to generate the output for the following instructions? Provide a justification for your answer.  
LW X5, 0(X5)  
SD X5, 0(X5) [3]
  3. Consider the following instruction mix of a code snippet given below. For a single-cycle Datapath scenario, answer the following questions:

| R-type | I-type<br>(non-lr) | Load | Store | Branch | Jump |
|--------|--------------------|------|-------|--------|------|
| 15%    | 20%                | 20%  | 10%   | 25%    | 10%  |

- a. Determine the fraction of all instructions where register write signals are enabled [1.5]
- b. Determine the fraction of all instructions where the final output come directly from the ALU [1.5]
- c. Determine the fraction of all instructions where the processor needs to perform sign extension [2]

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Duration : 20 Minutes

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**ID:**

**Name:**

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1. Draw a simplified datapath with control unit that can process *only* the following codes:  
LD X25, 16(X6)  
SD X6, 24(X25) [12]
  2. Does the ALU control utilize instruction bits 30 and 14-12 to generate the output for the following instructions? Provide a justification for your answer.  
LD X2, 0(X0)  
SH X4, 0(X2) [3]
  3. Consider the following instruction mix of a code snippet given below. For a single-cycle Datapath scenario, answer the following questions:

| R-type | I-type<br>(non-ld) | Load | Store | Branch | Jump |
|--------|--------------------|------|-------|--------|------|
| 10%    | 25%                | 15%  | 15%   | 20%    | 15%  |

- a. Determine the fraction of all instructions where register write signals are enabled [1.5]
- b. Determine the fraction of all instructions where the final output come directly from the ALU [1.5]
- c. Determine the fraction of all instructions where the processor needs to perform sign extension [2]